

RAJEEV KUMAR

B.Tech CSE (Artificial Intelligence & Data Engineering)

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EDUCATION

B.Tech in Computer Science & Engineering (AI & Data Engineering)

Lovely Professional University | Expected Graduation: 2028

PROJECTS

IntelliDocs-AI · AI-Powered Document Q&A System (RAG)

June · July 2026

Python · PyMuPDF · Gemini Embeddings · ChromaDB · Rank-BM25 · FastAPI · Streamlit · React

- Engineered an end-to-end RAG Document Q&A platform using Python, PyMuPDF, Google Gemini API (gemini-embedding-001), ChromaDB, and Rank-BM25, enabling hybrid vector & sparse semantic search over PDF documents with chunk-level source citations.
- Implemented Reciprocal Rank Fusion (RRF k=60) & resilient LLM fallback architecture cascading across Google Gemini model tiers (gemini-3.1-flash-lite -> gemini-2.0-flash -> gemini-1.5-flash -> mock) with automated retry logic, achieving 100% service uptime during API rate-limit spikes.
- Built a multi-document comparative analysis engine & dual user interface in Streamlit and React (Vite + TailwindCSS) backed by a FastAPI REST API, supporting side-by-side cross-document topic comparison and structural summarization across PDF corpora.

TECHNICAL SKILLS

Languages:	Python, SQL, JavaScript
ML & NLP:	Sentence Transformers (all-MiniLM-L6-v2), PyTorch (intro), Scikit-learn
LLM & GenAI:	Google Gemini API, RAG Architecture, Hybrid Search (Dense + BM25 RRF), Prompt Eng.
Databases:	ChromaDB (Local HNSW), PostgreSQL
Web & Backend:	FastAPI, Uvicorn, Streamlit, React, Vite, TailwindCSS
Tools & Testing:	Git, GitHub, Docker (intro), Pytest (10 test suites passed)
Concepts:	Retrieval-Augmented Generation, Vector Embeddings, Cosine Similarity, Reciprocal Rank Fusion

CERTIFICATIONS &

Summer Internship 2026 · Foundations of Applied Machine Learning

Lovely Professional University | June · July 2026

Problem Statement: I2 · Document Q&A (RAG over a Focused Corpus)